

Deep Learning-Based Real-Time Sign Language Translation System for Telehealth Applications

Problem Addressed

Communication between hearing- and speech-impaired individuals and healthcare professionals is difficult, especially in urgent medical situations. Traditional sign language recognition systems mainly focus on simple gesture recognition and lack real-time sentence-level translation required in telehealth environments.

Research Gap

Previous studies show several limitations in sign language translation systems:

- Most systems rely on **American Sign Language datasets**, which limits localization for regional sign languages.
- Many models focus on **isolated word recognition instead of sentence-level translation**.
- Lack of integration of **multiple motion features** such as skeleton, optical flow, and spatial-temporal cues.
- Limited implementation in **real-time telehealth applications**.

Key Contribution

The research proposes a **deep learning-based real-time sign language translation system** designed for healthcare communication in Sri Lanka. The system translates sign language gestures into text and supports real-time interaction between patients and doctors using socket-based communication.

Methodology

▪ Data Acquisition

Video input of sign language gestures is captured using a camera or mobile device. The system processes the video frames for gesture recognition.

▪ **Preprocessing**

Several preprocessing techniques are applied:

- Background removal using OpenCV
- Skeleton extraction using MediaPipe
- Optical flow calculation to capture motion information

These steps improve gesture detection accuracy.

▪ **Feature Extraction**

The system uses **Inflated 3D Convolutional Neural Network (I3D)** to extract spatial-temporal features from sign language videos. This model captures both appearance and motion patterns of gestures.

▪ **Sequence Modeling**

Three deep learning models were evaluated:

- **T5 Decoder model**
- **LSTM-based sequence model**
- **Transformer-based model**

Among them, the **Encoder + Transformer + Decoder architecture** achieved the best performance.

Dataset / Gesture Vocabulary

Three datasets were considered:

- **BBC-Oxford BSL Dataset**
- **RWTH-PHOENIX-Weather 2014T Dataset**
- **HOW2SIGN Dataset**

The **HOW2SIGN dataset** was selected because it contains large-scale multimodal sign language data and supports sentence-level translation tasks.

Results

Experiments showed that the **Transformer-based architecture** provided the best translation performance.

According to the results table in the paper:

Model	Test BLEU Score
Encoder + T5 Decoder	0.0129
Encoder + LSTM Decoder	0.0312
Encoder + Transformer Decoder	0.0478

The Transformer model captured temporal dependencies better and produced more accurate sentence translations.

Significance

The proposed system demonstrates the potential of AI-based sign language translation for healthcare applications. It enables real-time communication between deaf patients and medical professionals, improving accessibility and social inclusion

Future Work

Future improvements suggested include:

- Expanding datasets with **localized Sri Lankan Sign Language data**
- Improving translation accuracy and system latency
- Adding **sign-language avatars** to convert speech responses into visual gestures

These improvements would enhance communication between deaf individuals and healthcare providers.